

PianoKeystroke-EMG: Piano Hand Muscle Electromyography Estimation from Easily Accessible Piano Keystroke

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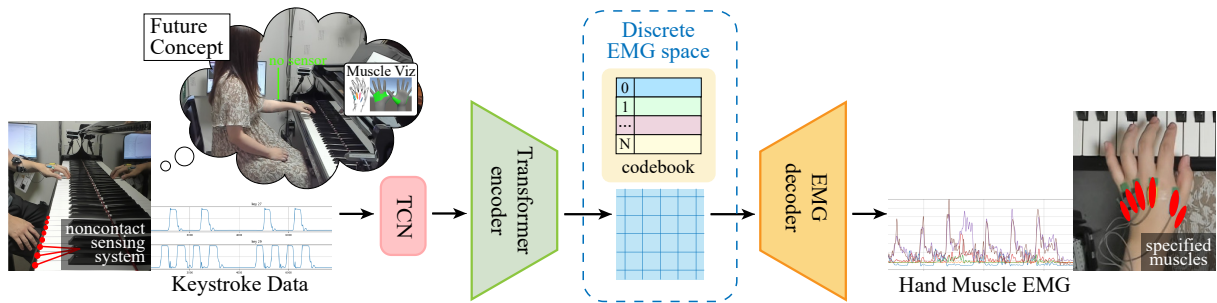


Figure 1: Overall flow of the proposed system. The pianist’s hand muscle electromyography (EMG) is estimated by leveraging accessible keystroke motions. The future concept includes an example of the potential application of this cost-effective pipeline.

Abstract

The acquisition of motor skills is intrinsically related to human movement dynamics. Traditional methods for measuring muscle activation via electromyography (EMG) sensors are often prohibitively expensive. Previous approaches developed algorithms to infer muscular information from other modalities, though predominantly concentrating on gross motor activities. In contrast, dexterous skills, such as piano performance, mostly engage small muscles, for which reliable and cost-effective solutions remain underexplored. We introduce *PianoKeystroke-EMG*, an approach to estimate EMG data from readily accessible data, such as piano keystroke motions. Further evaluation demonstrates that our system outperforms baselines in accuracy and versatility, potentially offering a practical tool for pianists to enhance their performance techniques.

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1 Introduction

Evaluating human movement dynamics is crucial for understanding muscle recruitment patterns and biomechanics in skill acquisition. Quantitative analysis of these dynamics can provide deeper insights than visual assessments, among which electromyography (EMG) is essential for recording muscle activations. However, costly sensors are required for measurement, which are resource-intensive and limit the application for large-scale studies and general use.

To address this, various methods have been developed to estimate muscle forces from more accessible modalities. Uhlich et al. utilized physics-based musculoskeletal simulations to compute muscle activations from videos, which proved effective for analyzing gross motor activities using large muscles [Uhlich et al. 2022]. Conversely, dexterous skills such as piano performance involve small hand muscles, which are vital for joint stability and manual dexterity. However, little has been unveiled about how to obtain accurate EMG data for small muscles at a low cost.

The present study introduces the concept of multimodal piano datasets (EMG, keystroke, and other modalities in the future) and explores how to estimate hand muscles’ EMG from cost-effective keystroke motions. Based on a noncontact sensing system [Liu et al. 2023], we propose *PianoKeystroke-EMG* that utilizes codebooks to infer precise EMG from keystrokes. Moreover, our system refines a pre-trained model with a minimal quantity of newcomer data, thereby facilitating adaptations to user-specific models. The evaluations demonstrate that this approach achieves accurate EMG estimation and exhibits robustness across diverse individuals.

2 Method

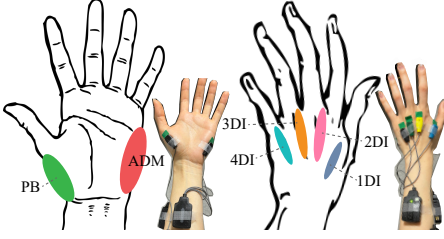


Figure 2: The six specified muscles of the left hand.

Data Collection. As a preliminary investigation for the proposed pipeline, we collect data from two participants (1 female and 1 male) with an average of 13 years of piano experience. The six hand muscles and four tasks are determined based on a professional pianist’s suggestions. Figure 2 illustrates the left-hand muscles: Abductor Digiti Minimi (ADM), Pollicis Brevis (PB), First (1DI), Second (2DI), Third (3DI), and Fourth Dorsal Interosseus (4DI). One of the participants repeated each task fifty times and the other ten times, with 2000 Hz EMG and 1000 Hz keystrokes. In total, the dataset comprises approximately 1.5 million frames of valid data.

Preprocess. We apply a Butterworth Filter (5th-order low-pass, 12 Hz cutoff frequency) to denoise the raw EMG. Then EMG signals are downsampled to 1000 Hz and normalized to their respective Maximal Voluntary Contraction (MVC) values. For data partitions, 25% of data from one participant is randomly reserved for model evaluation. To assess our model’s generalizability, Task 4 is exclusively included in the test set. Additionally, to evaluate the effectiveness of the few-shot customization framework, the other participant’s data is treated as unseen and excluded from the training process.

Discrete EMG Space. The ultimate goal is to estimate EMG from readily accessible keystroke data. To estimate EMG, incorporating EMG priors is essential to mitigate uncertainty and complement details. Previous research highlighted the benefits of a discrete codebook to ensure high-fidelity results with severely degraded input [van den Oord et al. 2018]. Motivated by this, we learn discrete latent representations of EMG by training a Vector Quantised-Variational AutoEncoder (VQ-VAE). Subsequently, a transformer encoder is trained to map the keystroke input to the corresponding EMG codebook. The encoded keystrokes are then decoded into EMG signals using the pre-trained VQ-VAE decoder, as depicted in Figure 1.

Few-shot Fine-tuning Architecture. Due to the divergence in EMG distributions across individuals, training personalized models for newcomers is necessary but computationally expensive and labor-intensive. To address the challenge, we employ Low-Rank Adaptation (LoRA) [Hu et al. 2021] to learn keystroke representations across different individuals. The codebook is frozen and the model’s linear connection layers are fine-tuned with minimal samples, hence facilitating few-shot fine-tuning and adaptation to various users.

3 Evaluation

For comparisons, since our work is the first to estimate EMG using keystrokes, we deploy a sequence-to-sequence transformer as

Table 1: Comparison results. Mean Squared Errors (MSE) of estimated EMG are calculated.

Tasks	Baseline (transformer)		Ours (codebook)	
	Avg.	Std.	Avg.	Std.
Task 1, 2, 3	0.075	± 0.046	0.017	± 0.017
Task 4	0.184	± 0.027	0.008	± 0.007



Figure 3: Some visualization results between baseline and our method. (A) Ground truth EMG. (B) Estimated EMG.

the baseline. Table 1 indicates that our codebook-based model outperforms the baseline for three tasks ($\text{MSE}=0.017\pm 0.017$) and Task 4 ($\text{MSE}=0.008\pm 0.007$), which is exclusively included in the evaluation. And Figure 3 presents some visualization results. On the other hand, our few-shot fine-tuning can achieve a 75% reduction in MSE of unseen data ($\text{MSE}=0.040\pm 0.030$) and markedly surpass the performance of the untuned baseline ($\text{MSE}=0.160\pm 0.065$).

4 Limitations and Future Work

This work explores multimodal piano datasets and proposes an EMG estimation method, *PianoKeystroke-EMG*, utilizing readily accessible keystroke motions, while several limitations remain for future investigation.

The current dataset is limited in scale and modalities. We plan to collect more pianists’ performance data in more modalities, including hand postures, audio, video from multiple perspectives, etc. With this large-scale dataset, economical data synthesis and inferences across modalities are possible. Thus it allows ordinary users to better understand muscle recruitment patterns in piano acquisition and facilitates other applications such as muscular disorders prevention (Figure 1: Future Concept).

Acknowledgments

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